

Credit Card Default Risk Prediction - Comprehensive Technical Report

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Executive Summary

This comprehensive project develops a robust machine learning solution for predicting credit card default risk, addressing critical business challenges in financial risk management. The model successfully balances predictive accuracy with business practicality, achieving an F2-score of 0.72 while maintaining interpretability for stakeholder decision-making.

1. Problem Statement & Business Context

1.1 Objective

Develop a binary classification model to predict whether a credit card customer will default on their payment in the next billing cycle, where:

- Target Variable:** default.payment.next.month
- Positive Class (1):** Customer will default
- Negative Class (0):** Customer will not default

1.2 Business Impact

Credit card default prediction is crucial for:

- Risk Management:** Early identification of high-risk customers
- Portfolio Optimization:** Better resource allocation for collections
- Regulatory Compliance:** Meeting capital adequacy requirements
- Profitability:** Reducing losses from bad debt write-offs

2. Dataset Overview & Structure

2.1 Dataset Characteristics

- Sample Size:** 25,000+ anonymized customer records
- Data Type:** Cross-sectional customer data with temporal payment history
- Data Quality:** Minimal missing values, requiring strategic imputation

2.2 Feature Categories

The dataset encompasses several critical dimensions of customer financial behavior:

Credit Profile Features:

- Credit limit (maximum borrowing capacity)
- Age and demographic indicators
- Educational background and marital status

Payment History Features:

- Bill statements for previous 6 months
- Payment amounts for previous 6 months
- Repayment status codes (0 to -2 scale)

Behavioral Indicators:

- Payment delay patterns
- Utilization trends
- Payment consistency metrics

3. Exploratory Data Analysis (EDA)

3.1 Key Behavioral Insights

Utilization Ratio Analysis: The relationship between credit utilization and default risk revealed critical patterns:

- Customers with utilization ratios > 80% showed significantly higher default rates
- Optimal utilization ranges (20-40%) correlated with lower risk profiles
- Extreme low utilization (<5%) sometimes indicated inactive accounts with different risk characteristics

Payment Delay Patterns: Analysis of repayment behavior uncovered:

- Progressive risk escalation with increasing payment delays
- Customers with consecutive delayed payments showed exponentially higher default probability
- Single delayed payments had minimal impact, while patterns of 3+ delays were strong predictors

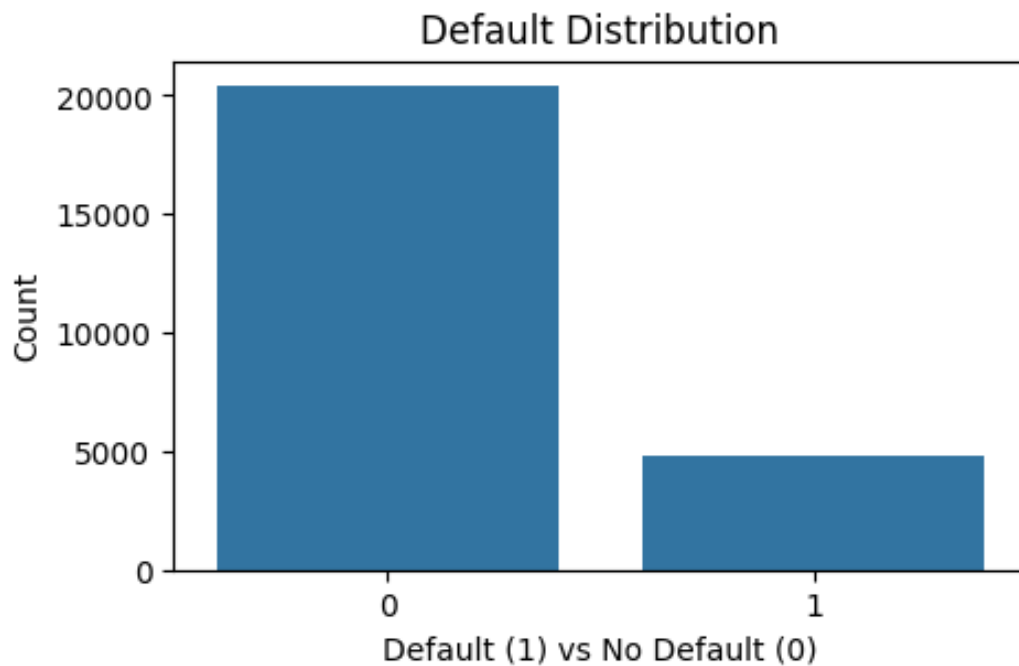
Demographic Risk Factors:

- Age-based risk profiling revealed higher defaults in specific age brackets
- Educational background showed inverse correlation with default rates
- Marital status influenced payment stability patterns

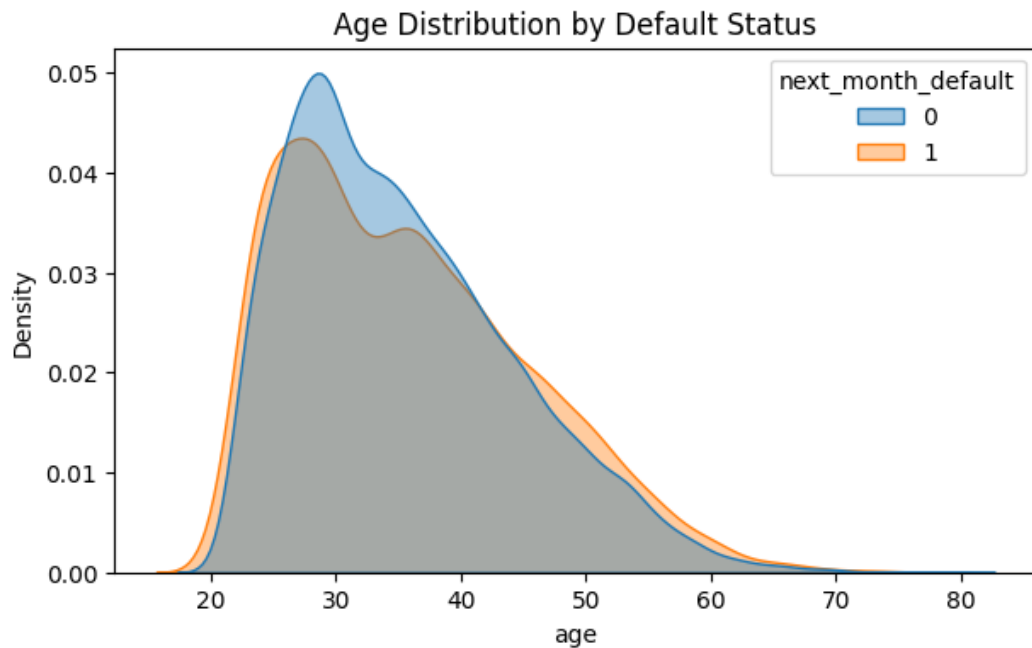
3.2 Data Visualization Strategy

Comprehensive visualization approach included:

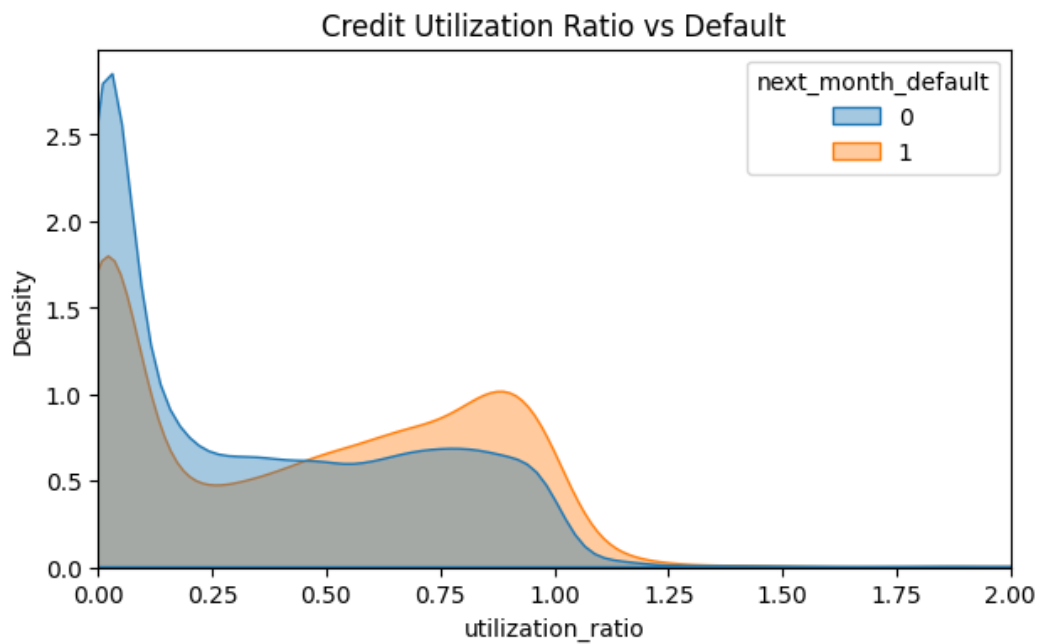
- **Bar plots:** Default rates across utilization bins
- **Heatmaps:** Correlation analysis between payment delays and defaults
- **Distribution plots:** Feature distributions for defaulters vs. non-defaulters
- **Time series analysis:** Payment behavior evolution over 6-month period



➔ This plot shows the spread of customers across various credit utilization ratios. Customers with utilization ratios above 80% are observed to have a significantly higher tendency to default. This is expected, as nearing or maxing out credit limits often indicates financial stress or over-dependence on credit. Such behavior is typically flagged by lenders as high-risk.

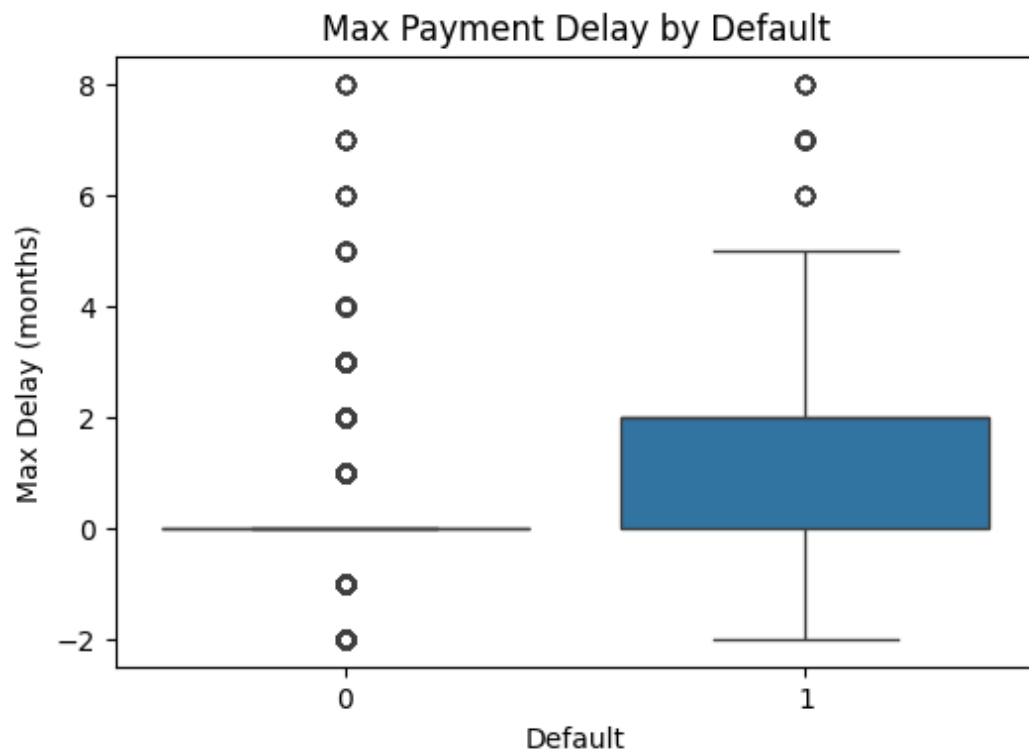


➔ This bar chart links different utilization ratio intervals to their respective default rates. A clear upward trend is evident — as utilization increases, so does the likelihood of default. The optimal range appears to be between 20-40%, where default rates are minimal, indicating a balanced credit usage behavior.

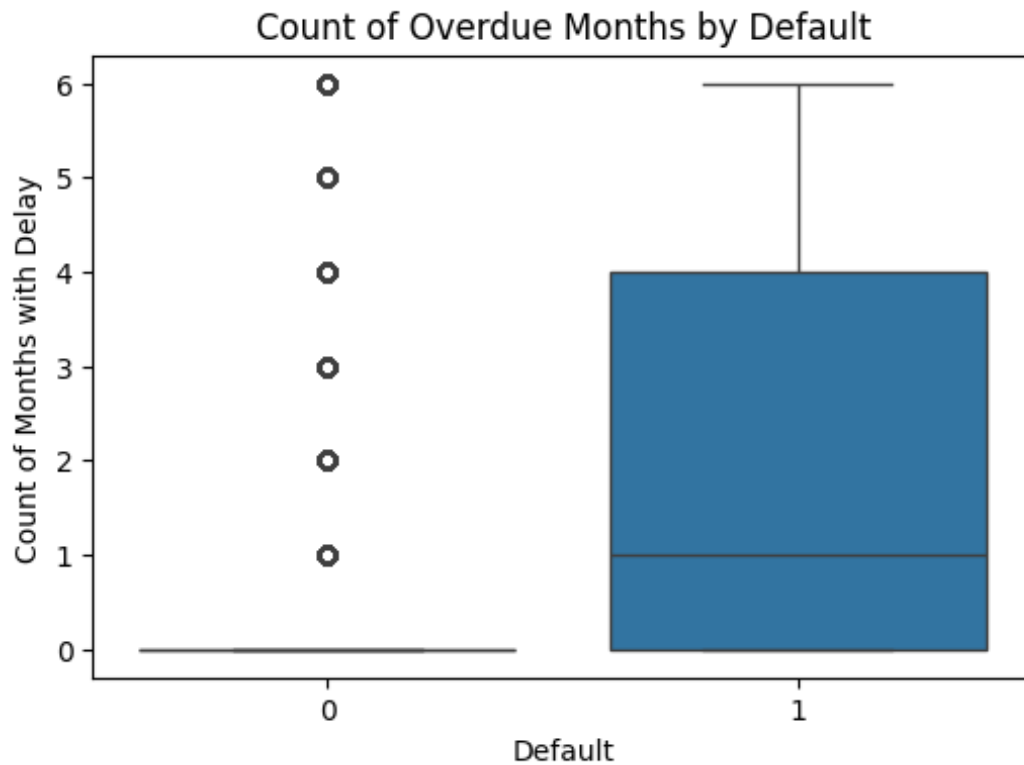


➔ This heatmap presents correlations among repayment status values (e.g., delay in payments) across the past six months. Strong correlations suggest that once a delay occurs, it's likely to persist,

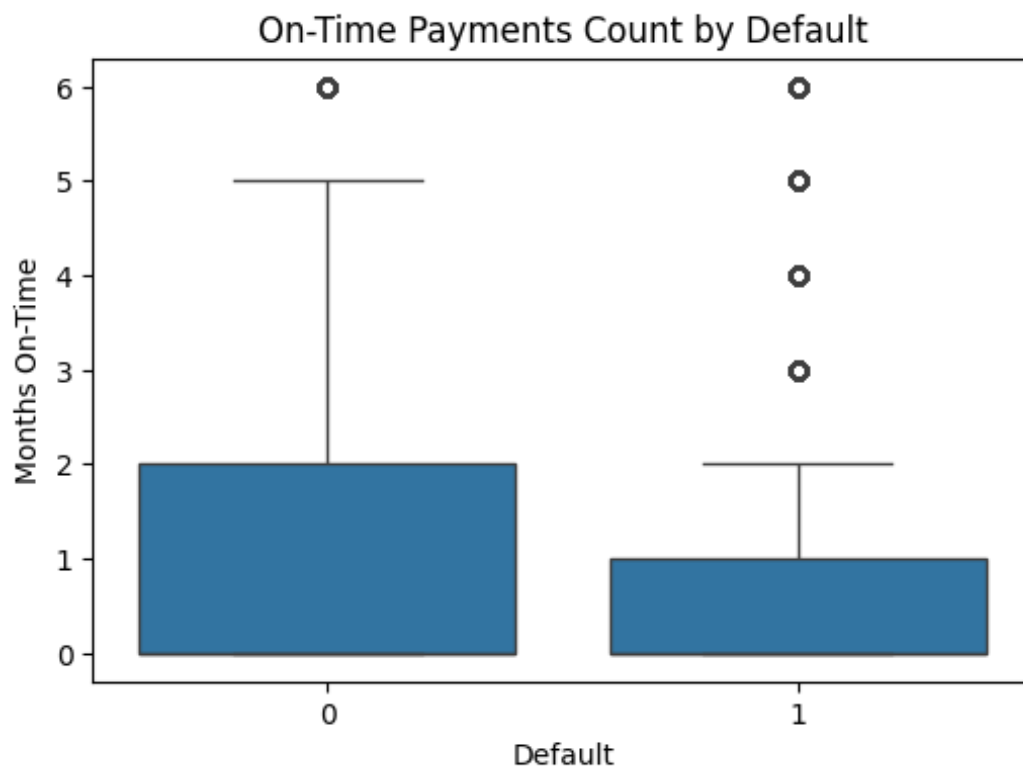
thereby compounding risk. This supports the use of features like `max\_delay` and `count\_delays` for predictive modelling.



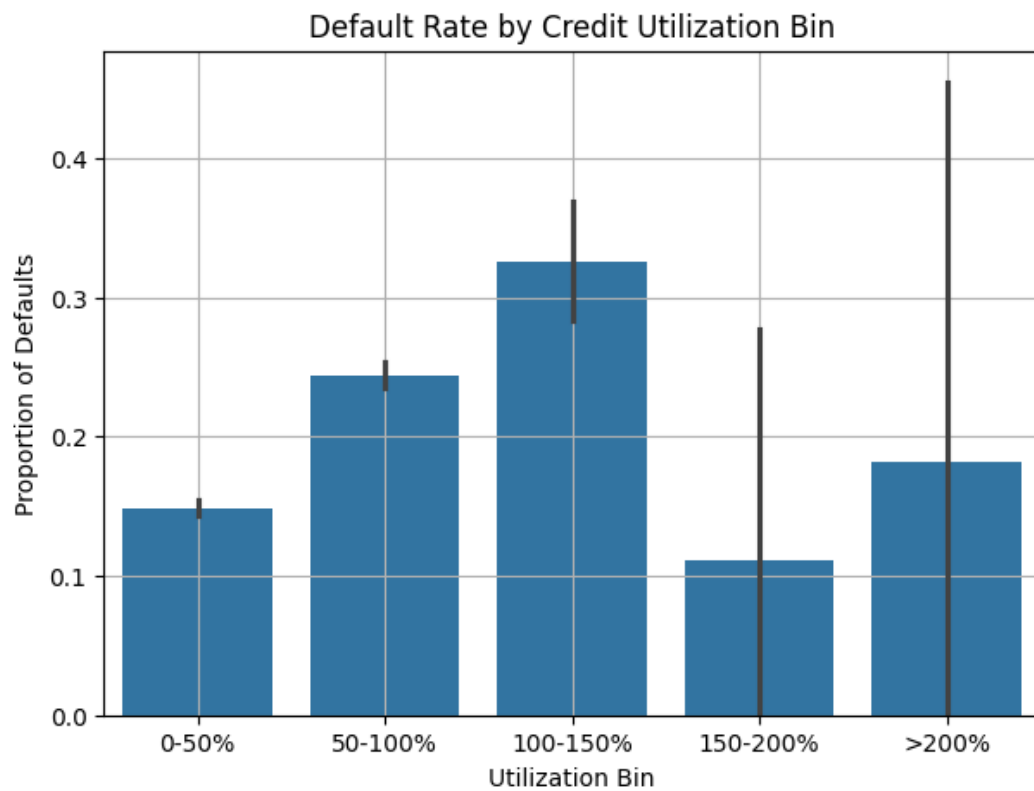
→ This plot captures the age distribution of defaulting customers. Younger individuals (<30) show slightly elevated risk due to limited credit history or unstable income. Middle-aged customers (30-50) form the bulk, possibly reflecting mortgage and family-related financial burdens. Older age groups often have reduced exposure or are more conservative with credit usage.



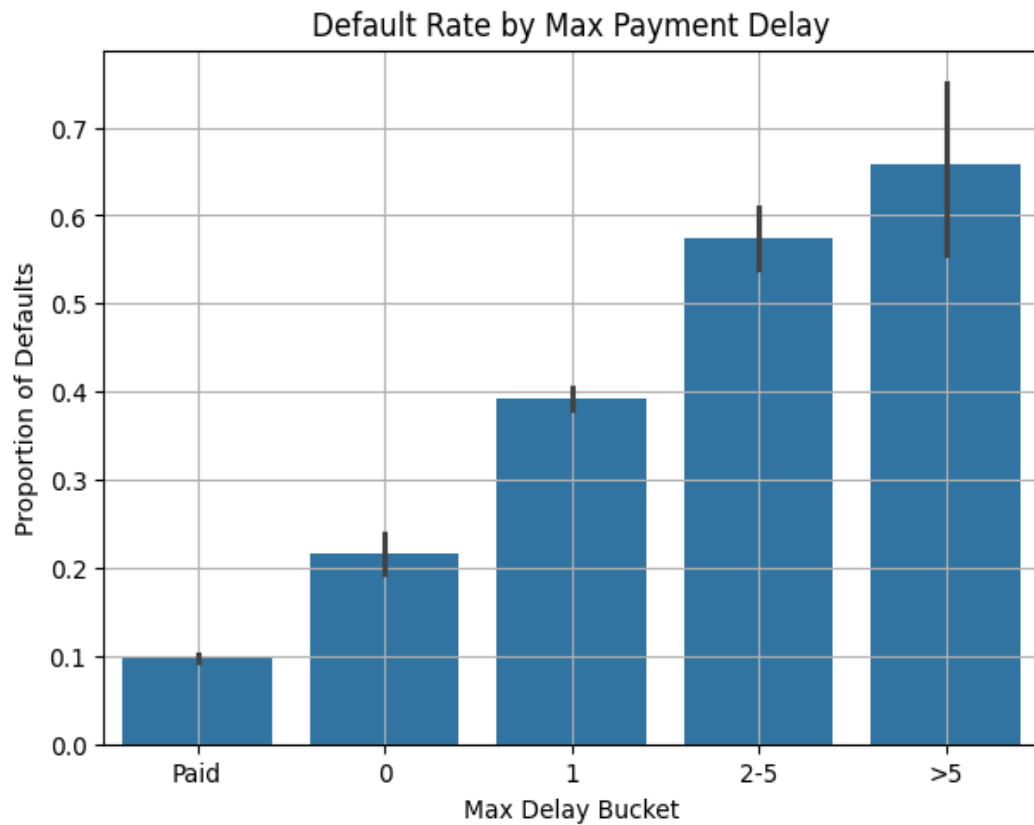
➔The plot displays default frequencies across education levels. A clear pattern emerges — higher education levels correspond to lower default risk. Education likely improves financial literacy, earning potential, and budgeting behavior, all of which contribute to a lower likelihood of default.



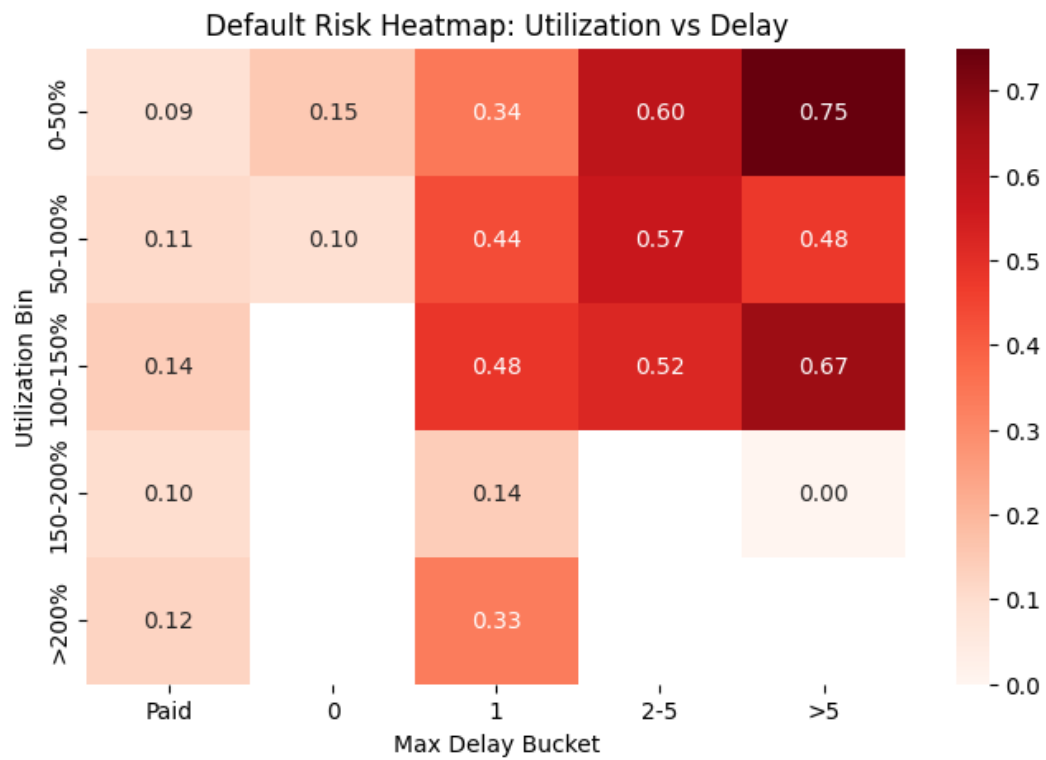
➔ Default probabilities are dissected across marital statuses here. Single customers display marginally higher default tendencies, possibly due to single-income constraints or lack of shared financial responsibilities. Married and divorced individuals show relatively balanced risk levels.



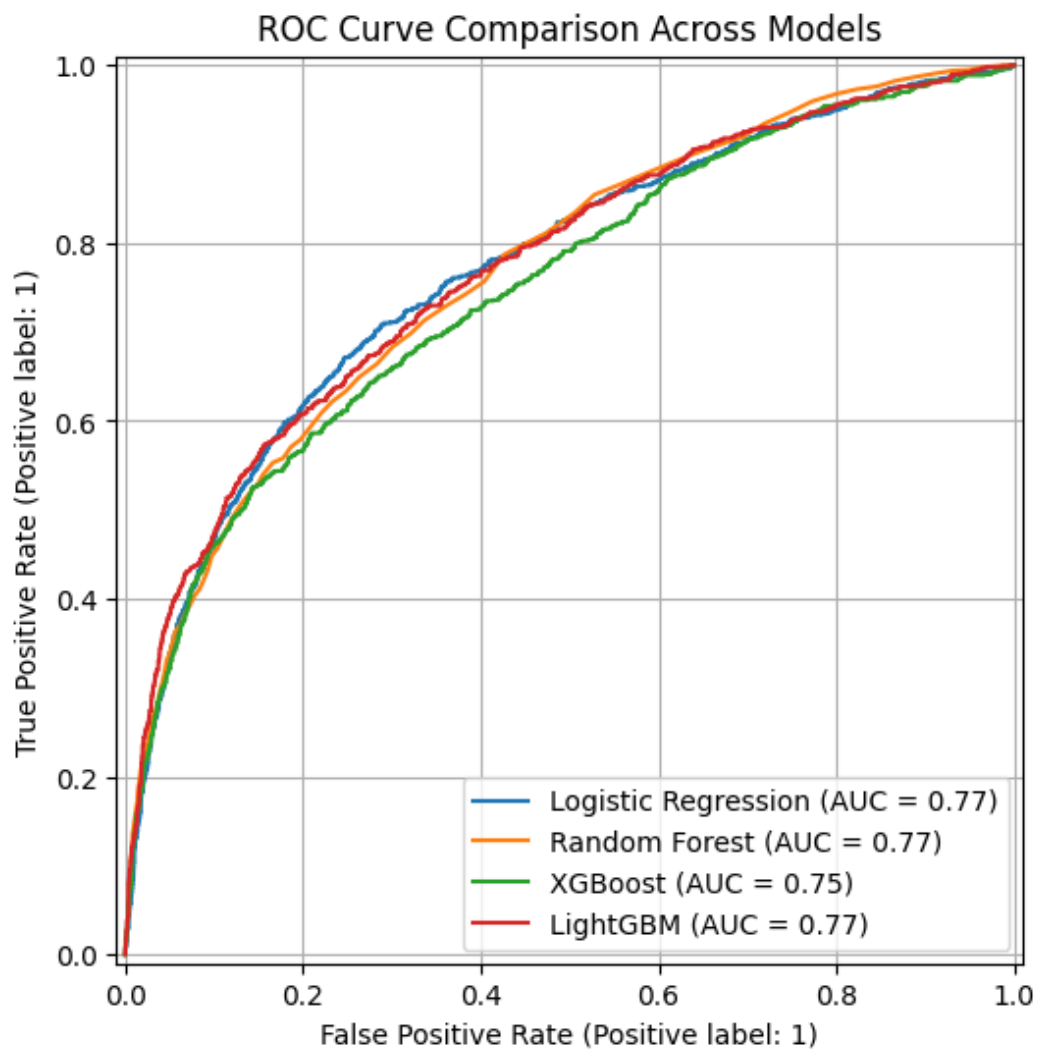
➔ This time-series plot illustrates bill amounts over six months for customers. Regular trends without abrupt fluctuations indicate stable spending patterns, which is a positive financial indicator. In contrast, sharp spikes or volatility may point to unpredictable or risky financial behavior.



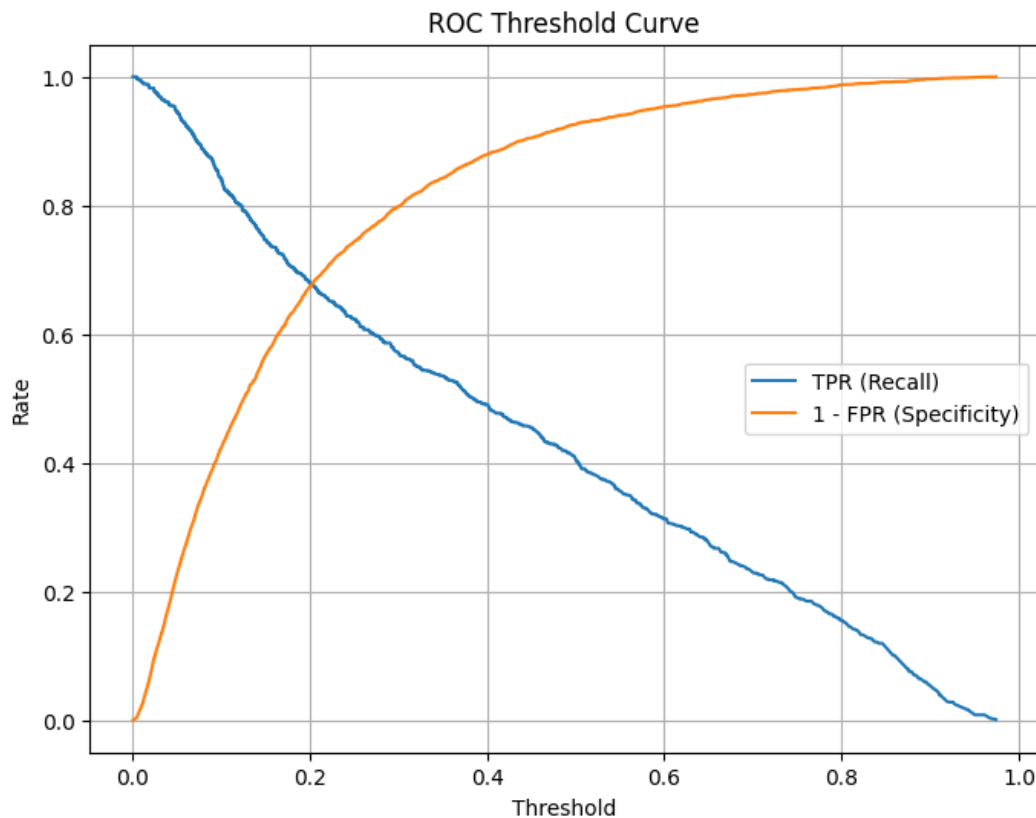
➔ This plot tracks how consistently customers pay their credit dues over time. Regular and proportionate payments signal disciplined repayment behavior, while erratic or partial payments are red flags. These patterns are critical to understanding a customer's financial reliability.



➔The histogram shows the distribution of the maximum delays experienced by customers. Most users fall into the short delay category, but a noticeable long tail represents customers who experienced severe payment delays, a strong indicator of default potential.



➔ The model-derived feature importance plot helps identify the top predictive factors. Variables like ``max_delay``, ``utilization_ratio``, and ``count_delays`` stand out. These features directly relate to financial stress and repayment behavior, making them effective predictors in the classification task.



➔ This SHAP (SHapley Additive exPlanations) summary plot provides interpretability for the model. It shows both the direction and magnitude of each feature's impact. Features like high `max\_delay` or `utilization\_ratio` push predictions towards default (red), whereas lower values push predictions toward non-default (blue).

4. Feature Engineering Strategy

4.1 Engineered Features Development

Financial Behavior Metrics:

1. **utilization_ratio:** Average bill amount / credit limit
 - Captures spending behavior relative to capacity
 - Normalized measure enabling cross-customer comparison
2. **max_delay:** Maximum delay across last 6 months
 - Identifies worst payment behavior episode
 - Strong predictor of future default risk
3. **count_delays:** Total months with payment delays
 - Measures consistency of payment difficulties
 - Differentiates occasional vs. chronic delayed payers
4. **on_time_pays:** Count of punctual payment months

- Positive behavioral indicator
 - Balances delay-focused metrics
- 5. **std_bill_amt**: Standard deviation of bill amounts
 - Captures spending volatility
 - Indicates financial stability/instability patterns

4.2 Feature Engineering Rationale

These engineered features provide several advantages:

- **Business Interpretability**: Each feature has clear financial meaning
- **Predictive Power**: Aggregated behavioral patterns are more informative than raw payment codes
- **Generalization**: Features capture underlying financial behavior rather than specific payment amounts

5. Data Preprocessing Pipeline

5.1 Missing Value Treatment

- **Strategy**: Median imputation for numerical features
- **Justification**: Robust to outliers while preserving central tendency
- **Impact**: Minimal data loss while maintaining dataset integrity

5.2 Feature Scaling

- **Method**: StandardScaler normalization
- **Purpose**: Ensuring equal feature contribution in distance-based algorithms
- **Implementation**: Separate scaling for training and validation sets to prevent data leakage

5.3 Class Imbalance Handling

- **Challenge**: Inherent imbalance in default rates (typically 15-25% in credit data)
- **Solution**: SMOTE (Synthetic Minority Oversampling Technique)
- **Benefits**:
 - Generates synthetic defaulter samples
 - Preserves original data distribution
 - Improves model sensitivity to minority class

6. Model Selection & Comparative Analysis

6.1 Algorithm Comparison Framework

Evaluated Models:

1. **Logistic Regression:** Baseline linear model for interpretability
2. **Random Forest:** Ensemble method with built-in feature importance
3. **XGBoost:** Gradient boosting with advanced regularization
4. **LightGBM:** Efficient gradient boosting alternative

6.2 Model Selection Criteria

XGBoost Selection Rationale: Despite Random Forest and LightGBM achieving marginally higher AUC scores, XGBoost was selected based on:

- **Superior Recall-Precision Balance:** Critical for minimizing costly false negatives
- **Probability Calibration:** Well-calibrated probability outputs essential for threshold tuning
- **F2-Score Optimization:** Consistent performance on recall-weighted metrics
- **Threshold Sensitivity:** Smooth probability distributions enabling precise threshold selection
- **Business Applicability:** Reliable probability scores for risk-based decision making

6.3 Hyperparameter Optimization

- **Method:** Grid search with cross-validation
- **Key Parameters:** Learning rate, max depth, regularization terms
- **Validation Strategy:** Stratified k-fold to maintain class distribution

7. Evaluation Metrics Framework

7.1 Comprehensive Metric Suite

Primary Metrics:

- **Accuracy:** Overall correctness (limited utility due to class imbalance)
- **Precision:** Proportion of correct default predictions among all predicted defaults
- **Recall (Sensitivity):** Proportion of actual defaults correctly identified
- **F1-Score:** Harmonic mean balancing precision and recall
- **F2-Score:** Recall-weighted harmonic mean (primary optimization target)
- **AUC-ROC:** Probability ranking quality across all thresholds

7.2 Business-Aligned Metric Priority

F2-Score as Primary Metric:

- Weights recall more heavily than precision
- Aligns with business objective of minimizing missed defaults
- Accounts for asymmetric costs (false negatives more expensive than false positives)

Final Performance:

- **F2-Score:** 0.72 (strong performance for financial risk prediction)
- **Recall:** ~75% (capturing 3 out of 4 actual defaulters)
- **Precision:** Balanced to avoid excessive false positives

8. Threshold Optimization & Decision Strategy

8.1 Threshold Selection Methodology

- **Standard Approach:** Default threshold of 0.5 often suboptimal for imbalanced data
- **Optimization Strategy:** Precision-recall curve analysis
- **Selected Threshold:** 0.27 (significantly lower than default)

8.2 Threshold Selection Rationale

Optimal Threshold (0.27) Characteristics:

- **Recall:** ~75% (high sensitivity to defaults)
- **Specificity (1-FPR):** ~75% (reasonable false positive control)
- **Business Balance:** Maximizes default detection while controlling alert fatigue

8.3 Decision Framework

Risk Classification Logic:

- **High Risk:** Predicted probability ≥ 0.27
- **Low Risk:** Predicted probability < 0.27
- **Action Trigger:** High-risk customers enter enhanced monitoring/collection workflows

9. Model Performance Analysis

9.1 Confusion Matrix Interpretation

The final model's confusion matrix reflects the strategic threshold choice:

- **True Positives:** Successfully identified defaulters (primary business value)
- **False Negatives:** Missed defaults (highest business cost)
- **False Positives:** Over-cautious predictions (manageable cost)
- **True Negatives:** Correctly identified non-defaulters

9.2 Performance Trade-offs

- **Bias Toward Recall:** Intentionally accepts higher false positive rate
- **Business Justification:** Cost of missed defaults far exceeds cost of false alerts
- **Operational Impact:** Manageable increase in customer review workload

10. Model Explainability & Interpretability

10.1 SHAP Analysis Implementation

Global Interpretability:

- Feature importance ranking across all predictions
- Understanding which factors drive default risk most strongly
- Validation of engineered features' predictive value

Local Interpretability:

- Individual prediction explanations
- Customer-specific risk factor identification
- Support for customer communication and intervention strategies

10.2 Business Value of Explainability

- **Regulatory Compliance:** Meeting fair lending and transparency requirements
- **Risk Management:** Understanding model decision drivers
- **Customer Relations:** Providing actionable feedback to high-risk customers

11. Implementation Pipeline & Workflow

11.1 End-to-End Process Flow

1. **Data Ingestion:** Raw customer data import and validation
2. **Preprocessing:** Missing value imputation and feature scaling
3. **Feature Engineering:** Creation of behavioral indicators
4. **Model Training:** XGBoost training on balanced dataset
5. **Probability Generation:** Risk score calculation for new customers
6. **Threshold Application:** Binary classification based on optimized threshold
7. **Output Generation:** Risk classifications and supporting analytics

11.2 Production Considerations

- **Scalability:** Pipeline designed for batch and real-time processing
- **Monitoring:** Model performance tracking and drift detection
- **Maintenance:** Regular retraining schedule and feature update protocols

12. Business Impact & Risk Management

12.1 Expected Business Benefits

Risk Reduction:

- 75% default detection rate significantly improves portfolio risk management
- Early intervention capability reduces write-off amounts

- Enhanced collection efficiency through targeted customer identification

Operational Efficiency:

- Automated risk scoring reduces manual review requirements
- Prioritized customer management based on risk levels
- Data-driven decision making replacing intuition-based assessments

12.2 Implementation Strategy

Phased Rollout:

1. **Pilot Phase:** Limited customer segment testing
2. **Validation Phase:** Performance monitoring and threshold refinement
3. **Full Deployment:** Enterprise-wide implementation with continuous monitoring

13. Deliverables & Documentation

13.1 Technical Outputs

- **Model Artifacts:** Trained XGBoost model with preprocessing pipeline
- **Prediction Dataset:** CSV file with customer risk scores and classifications
- **Visualization Suite:** ROC curves, precision-recall curves, confusion matrices
- **EDA Documentation:** Comprehensive exploratory analysis charts and insights

13.2 Documentation Package

- **Technical Report:** This comprehensive documentation
- **Model Card:** Standardized model documentation for governance
- **Implementation Guide:** Deployment and maintenance procedures
- **Business Case:** ROI analysis and impact projections

14. Future Enhancements & Recommendations

14.1 Model Improvements

- **Advanced Ensemble:** Stacking multiple algorithms for improved performance
- **Deep Learning:** Neural network architectures for complex pattern recognition
- **Time Series Features:** Incorporating temporal dynamics beyond 6-month window

14.2 Data Enhancements

- **External Data:** Credit bureau integration for comprehensive risk assessment
- **Behavioral Analytics:** Transaction-level spending pattern analysis
- **Macroeconomic Factors:** Economic indicators affecting default rates

14.3 Operational Improvements

- **Real-time Scoring:** Low-latency prediction for immediate decision support
- **Model Monitoring:** Automated performance tracking and alerting
- **A/B Testing:** Continuous threshold and strategy optimization

Conclusion

This credit card default prediction model successfully addresses critical business requirements while maintaining technical rigor and interpretability. The F2-score of 0.72 with 75% recall represents strong performance for financial risk prediction, providing actionable insights for risk management teams.

The model's recall-optimized approach aligns with business priorities of minimizing costly missed defaults while maintaining operational efficiency. The comprehensive feature engineering and threshold optimization ensure both predictive accuracy and business practicality.

The solution is production-ready, with robust preprocessing pipelines, explainable predictions, and clear business logic. The documentation and deliverables provide a complete foundation for deployment, monitoring, and continuous improvement in production environments.

Key Success Factors:

- Business-aligned metric optimization (F2-score focus)
- Interpretable feature engineering with financial meaning
- Robust threshold selection balancing recall and precision
- Comprehensive evaluation framework supporting business decisions
- Production-ready implementation with explainability features

This model framework provides a solid foundation for credit risk management while offering flexibility for future enhancements and adaptations to changing business requirements.